Jonathan Zhang

jonathantzhang.com | jzhan367@jh.edu | LinkedIn | GitHub

Education

Johns Hopkins University – GPA: 3.9 August 2023 – May 2027

- BS in Applied Mathematics and Statistics, BS in Biomedical Engineering, Minor in Computer Science
- Graduate level coursework: Machine Learning, Causal Inference, Probability, Honors Mathematical Statistics

Publications

- **Jonathan Zhang**, Erik Skalnes, Jacob Chen, Michael Oberst. "Bounding the Causal Impact of ML-assisted Decision-Making via Counterfactual Correctness." Accepted at UAI 2026. arXiv:2607.21806.

Research Experience

Machine Learning Researcher, Oberst Lab – Baltimore, MD March 2025 – Present

- Developed novel partial-identification bounds on the downstream causal impact of untrialed ML decision-support models without the need for randomized controlled trials.
- First-authored a paper on the findings with the professor and PhD students; accepted at UAI 2026 and presented at the 2025 JHU Research Symposium on Engineering in Healthcare.
- Currently evaluating computer-aided biopsy systems, developing methods to determine whether such systems improve biopsy outcomes despite an unobservable false negative rate.

Bioinformatics Researcher, Feinberg Lab – Baltimore, MD May 2025 – Present

- Analyzing whole-genome sequencing data with R and Julia on a remote Linux cluster.
- Developing an e-value approach to differential methylation region identification, reducing the false positive rate and improving power over Benjamini-Yekutieli adjusted p-values across millions of simultaneous hypothesis tests.
- Built a mediation analysis framework for a reduced-methylation diet study in mice, quantifying direct and indirect effects and showing that the mediating effect of appetite on fat gain was insignificant.
- Consolidated the lab's scattered analysis scripts into unified pipelines, cutting required user input by more than half and speeding up runtime by 15%.

Immunology Researcher, Baumgarth Lab – Baltimore, MD May 2024 – December 2024

- Proficient in handling mice, flow cytometry, cell culture, and western blot data collection and analysis.
- Experienced using R, MATLAB, and ImageJ to analyze and visualize collected data.

Teaching Experience

Teaching Assistant, Machine Learning & Probability – Baltimore, MD August 2024 – Present

- Teaches recitation and drafts homework for EN.601.475/675, a graduate level machine learning course.
- Leads a recitation section, holds office hours, and grades student work for EN.553.420/620: Probability, a graduate level probability course.
- Coordinates with professors to plan and develop review material for students.

Entrepreneurship & Leadership

COO/Co-founder, Aspire: the Pre-Med Planner – Baltimore, MD June 2024 – Present

- Developed a cross-platform mobile and web app in Flutter/Dart with a Firebase backend for pre-med students, growing to over 500 users.
- Created an AI-powered advising chatbot using the OpenAI API and engineered prompts to integrate user data into the agent context.
- Finalist at the SPARK and HopStart venture competitions, with $500 in total competition earnings.
- Learn more or download at aspiredoctor.com or the Apple App Store.

Founder, Parkinson's Pals at JHU – Baltimore, MD 2024 – Present

- Founded and lead a volunteer organization pairing undergraduate students with Parkinson's patients for companionship and support.
- Grew the organization to over 50 members and more than 20 patient pairs.
- Learn more at jhuparkinsonspals.jonathantzhang.com.

Projects

DeltaP: Bioimpedance Urine Output Monitor 1st Place, SPARK SP2026 ($2500)

- Built a bioimpedance-based urine output monitor for ICU patients to help prevent acute kidney injury, developed with a JHSOM professor and practicing nephrologist as part of the JHU BME Design Teams.
- Won 1st place at the SPARK SP2026 showcase ($2500) and presented at the JHU Design Day showcase.

Breast Cancer Identification App for Resident Physicians github.com/jonathanzhang58/MammologyApp

- Collaborating with a JHSOM professor and practicing radiologist to build a training app for over 50 radiology residents in breast cancer identification using IRB-approved de-identified patient mammograms.
- Developing a publicly accessible version; a MATLAB app prototype is available without patient data.
- Co-authoring a research paper on the app's impact on residents; finalist in the JHU CBID Healthcare Design Competition.

Multiple Sequence Alignment Using Monte Carlo Tree Search github.com/jonathanzhang58/MCTS_for_MSA

- Reframed multiple sequence alignment, a key input to protein structure prediction, as a search problem and solved it with Monte Carlo tree search methods (MCTS, NRPA, and a Beam NRPA variant).
- Benchmarked against classical star alignment and fed the results into AlphaFold2 to measure downstream protein-folding accuracy.

NASA C-MAPSS Predictive Engine Maintenance
github.com/jonathanzhang58/NASA-C-MAPPS-Predictive-Maintenance

- Built an engine failure pre-warning system predicting the remaining useful life of turbofan jet engines from 21-sensor time-series data in NASA's C-MAPSS dataset.
- Trained XGBoost models with engineered rolling-window features, proving over 4x more cost-effective than threshold-based baselines.

Kalman Filter Visualizer github.com/jonathanzhang58/Kalman-filter-algorithm-and-visualizer-

- Implemented the Kalman filter in MATLAB to predict unknown parameters from covariance with measured values.
- Presented findings at a departmental conference to over 50 students and professors as part of the JHU Directed Reading Program in AMS.

Home Server June 2024

- Transformed an old laptop into a home server running on Ubuntu Server OS.
- Configured the server with a VPN to the home network, enabling remote file storage and video streaming via ssh.
- Packaged the applications into Docker containers to optimize the server for compatibility.

Skills

Programming: Python, Java, MATLAB, R/R-studio, Julia, Flutter/Dart, UNIX/Linux, NumPy, TensorFlow
Laboratory: Flow cytometry, cell culture, western blot, mouse handling, ImageJ
Other: Firebase, Docker, OpenAI API, whole-genome sequencing analysis